# BACKWARD CHAINING

## AIM

To write a Prolog program to implement Backward Chaining by starting with a goal and working backward through rules to determine whether the goal can be proved.

## ALGORITHM

1. Start.
2. Define facts and rules.
3. Specify a goal to be proved.
4. Search for a rule whose conclusion matches the goal.
5. Treat the rule's conditions as sub-goals.
6. Recursively prove each sub-goal.
7. If all sub-goals are satisfied, the goal is true.
8. Otherwise, backtrack and try another rule.
9. Display the result.
10. Stop.

## SOURCE CODE

```
% Backward Chaining


fact(sunny).

fact(warm).


rule(pleasant, sunny).

rule(comfortable, warm).

rule(happy, pleasant).

rule(happy, comfortable).


prove(Goal) :-
```

```
    fact(Goal).

prove(Goal) :-
    rule(Goal, Condition),
    prove(Condition).
```

**Queries**

?- prove(pleasant).

true.

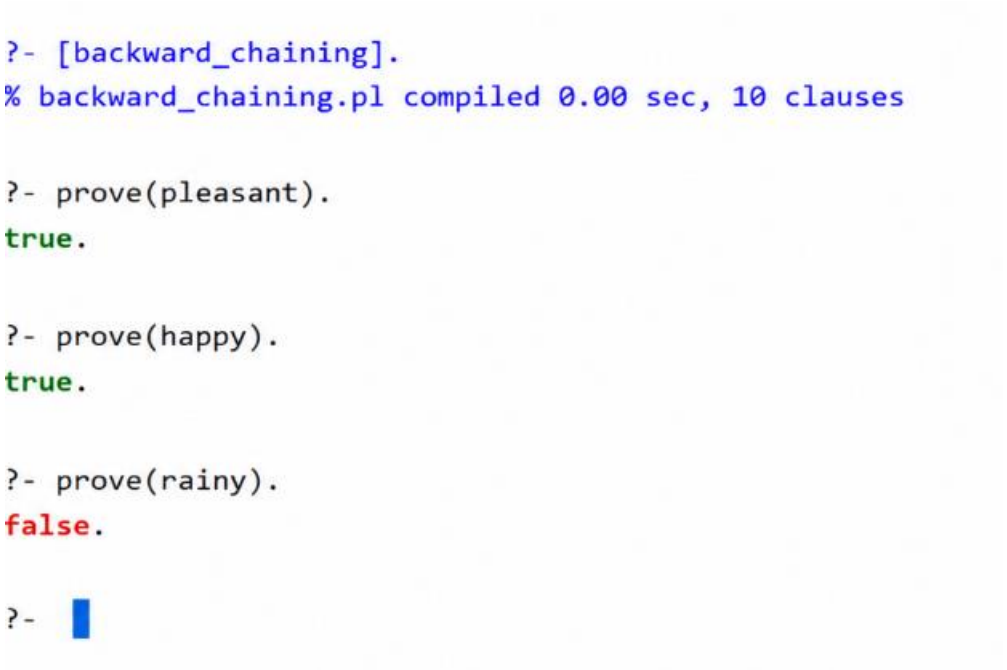
?- prove(happy).

true.

?- prove(rainy).

false.

**OUTPUT**




**RESULT**

The Prolog program successfully implements Backward Chaining by starting with the required goal and recursively checking whether the necessary conditions can be proved from the available facts and rules.